Magic Byte

Scenario Overview

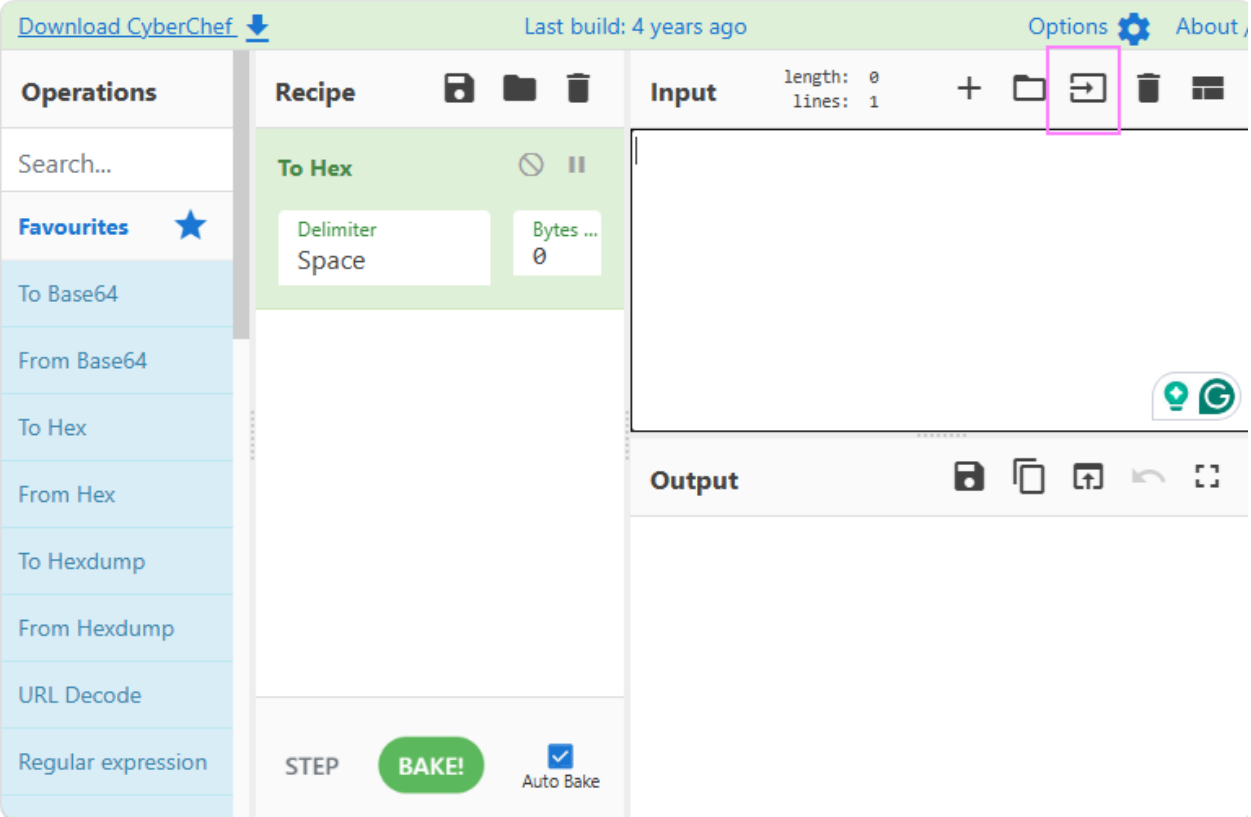
A suspicious image file was recovered by the security team, but the file appeared to have been altered or corrupted. Although it was labeled as a JPEG, initial inspection suggested that the file's internal structure did not match its extension. Your task is to analyze the file's magic bytes, determine its true file type, repair the corrupted header, and recover the original image to extract the hidden flag.

Tasks

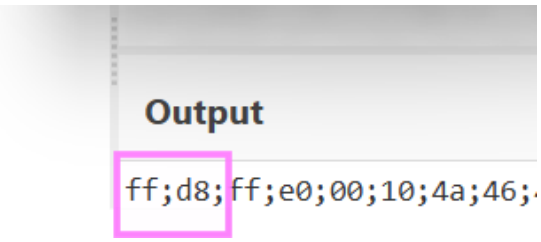
1. **Identify the true file type using magic bytes**
    - Upload the file to a tool such as CyberChef.
    - Inspect the first few bytes of the file to determine whether the header matches the claimed JPEG format.
    - Compare the observed magic bytes to known file signatures to identify the correct format.
2. **Confirm file structure using string analysis**
    - Run a strings recipe or similar analysis to detect internal markers (e.g., IDATx) that indicate the actual file type.
    - Validate whether the file is a PNG rather than a JPEG.
3. **Repair the corrupted file header**
    - Open the file in a hex editor (e.g., Hexedit).
    - Reference the correct PNG header structure.
    - Modify the magic bytes and any missing structural values to restore the file to a valid PNG format.
    - Save the corrected file and attempt to open it.
4. **Recover the hidden flag**
    - Once the repaired PNG opens successfully, inspect the image for the embedded flag.
    - Document the recovered flag as part of the final findings.

1. What is the original file type?

Use the online tool CyberChef by using the following link which alread has the Recipe that we need https://cyberchef.io/#recipe=To_Hex('Space',0) and upload your file by selecting the arrow on the right hand corner. The file provided for this lab was saved as “flag.jpeg”

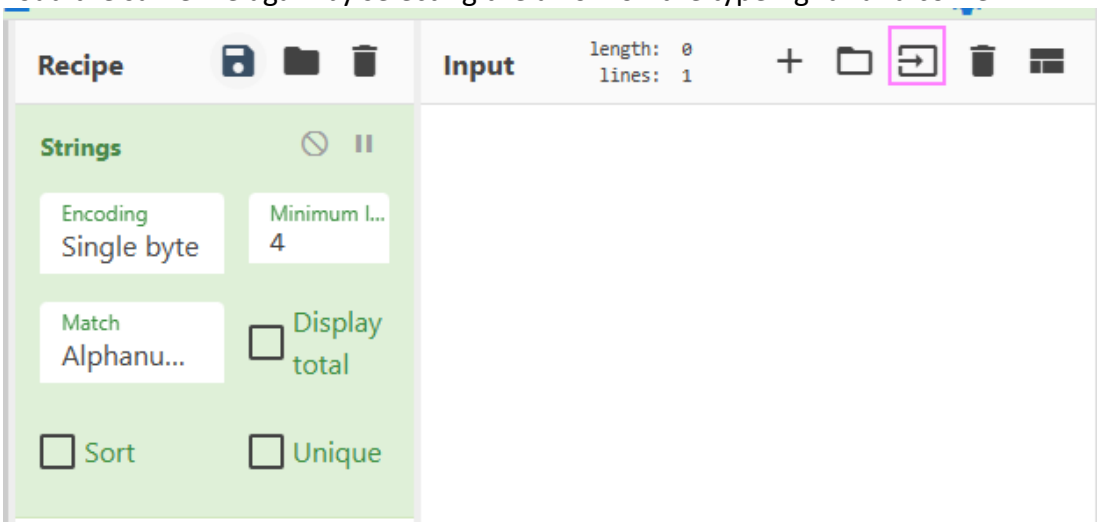


Once the file is loaded you will see your “data under the Output box. Here we are vieweing the first 4 magice bytes for jpeg listed “ff d8”



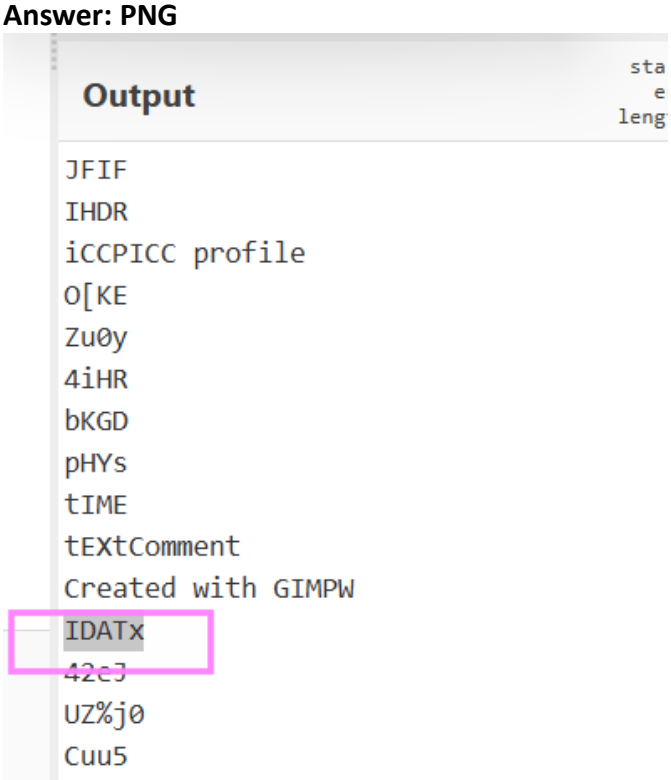
Next, navigate to the following CyberChef link with the pre selected String recipe for the following step:
https://cyberchef.io/#recipe=Strings('Single%20byte',4,'Alphanumeric%20%2B%20punctuation%20(A)',false,false,false)

Load the same file again by selecting the arrow on the type right hand corner.



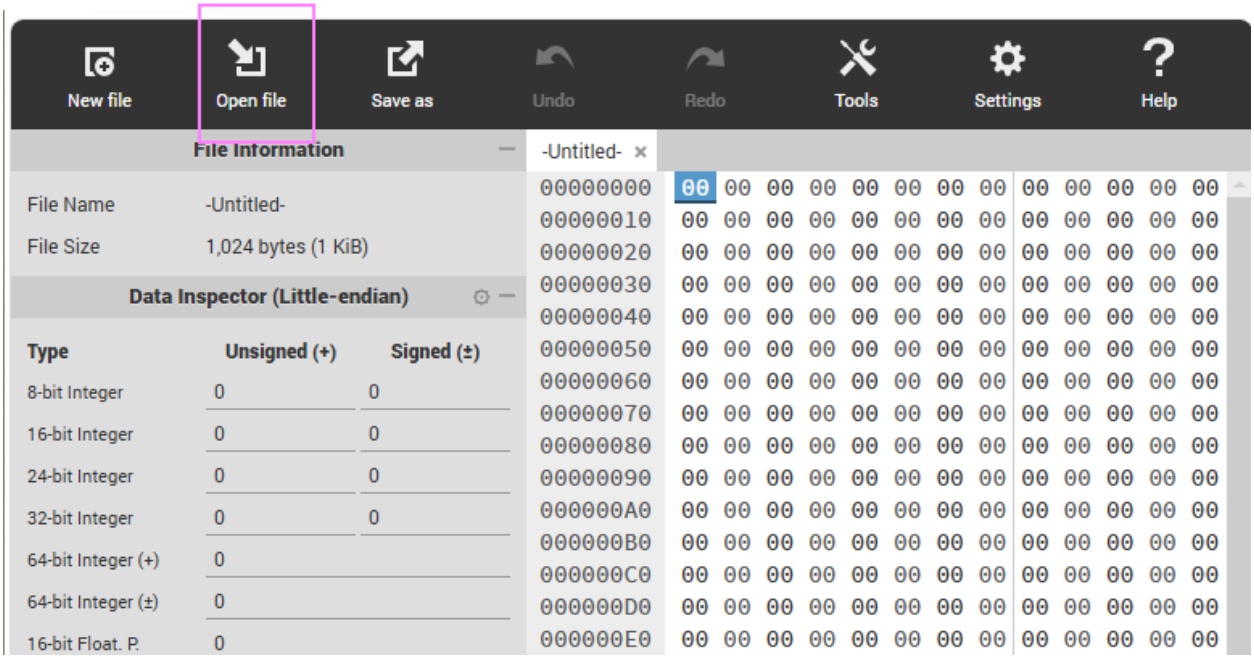
We then see listed “IDATx” which indicates that it is a PNG file not a JPEG file.

Answer: PNG

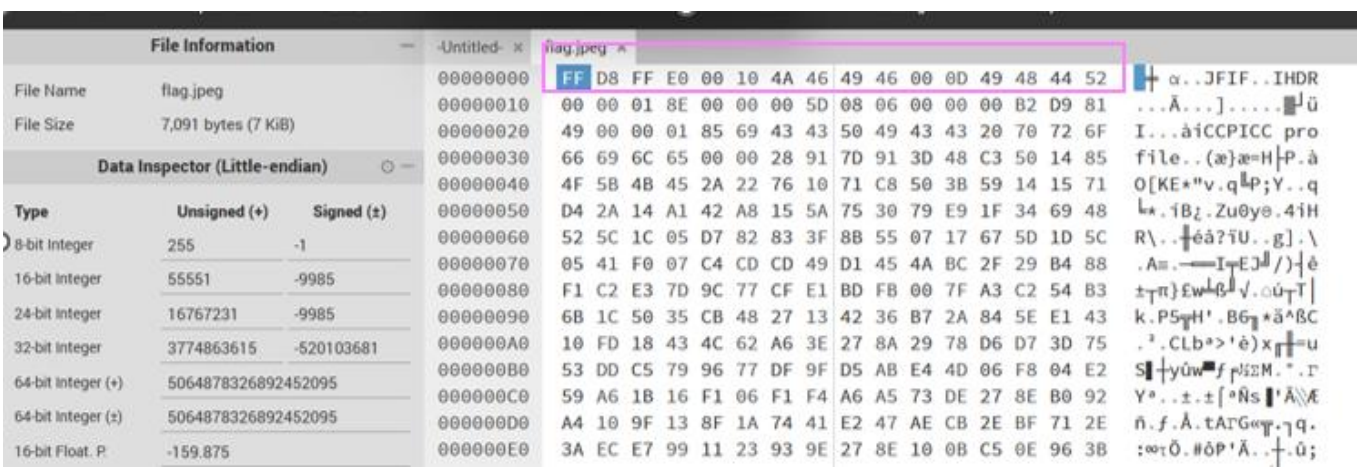


2. What is the flag?

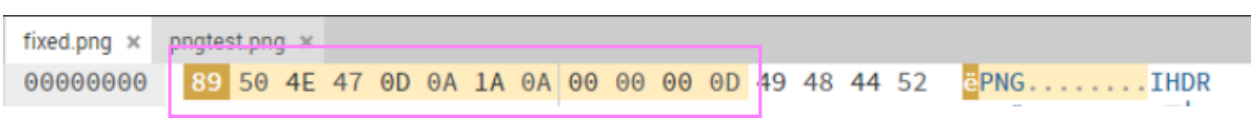
Navigate to the online tool named Hexedit, https://hexed.it and upload the file by selecting "Open file" on the top left corner. Here we are going to repair the corrupted header by inserting the correct magic bytes.



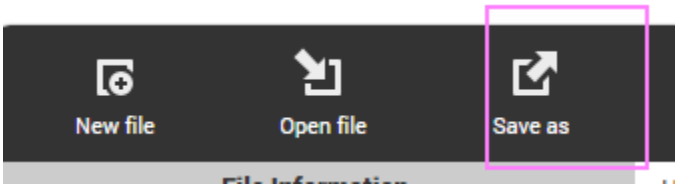
Here we see the current signature.

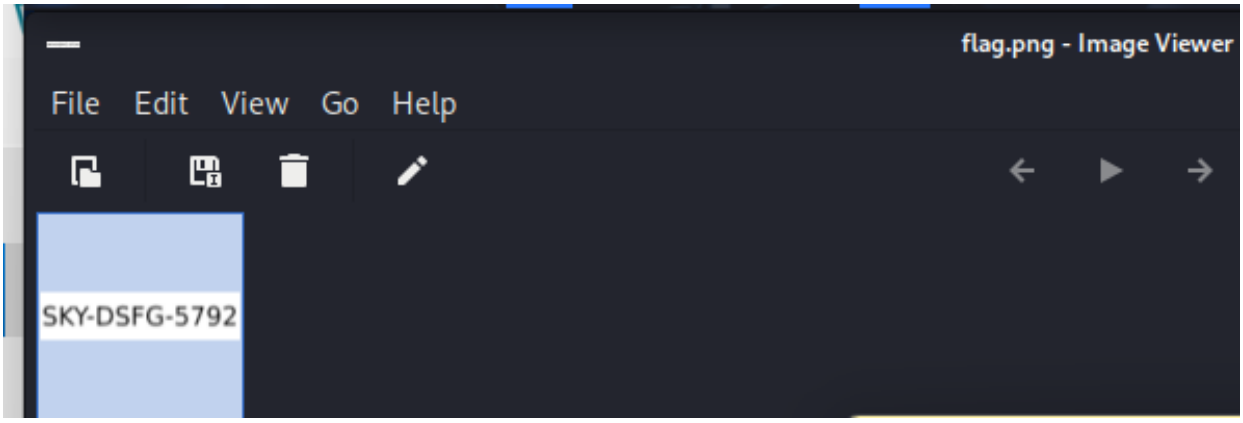


Next, we need to edit the raw file to replace the jpeg file signature with the PNG file signature 89 50 4E 47 0D 0A 1A 0A 00 00 00 0D



Resave the file as “flag.png” by clicking “Save as” on the top left corner.



Open the file and you should now be able to see the flag.

Answer: SKY-DSFG-5792